Motivation

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Motivation

Purpose of this chapter

The Motivation chapter explains why the SAF exists. It also explains why architecture decisions in an NSI are taken the way they are. It records the forces that act on statistical organisations, the results they want to achieve, the rules that guide their decisions, and the parties whose concerns must be addressed.

Without an explicit motivation, architecture is a set of models without a reason. With it, every capability, process, data state, application or platform choice can be traced back to a need and a result. That traceability makes architecture defensible, whether in a governance board, a budget discussion or an international review.

Position in the framework

Motivation is the first layer of the SAF and the foundation for the other layers. The Strategy layer translates motivation into capabilities and value streams. The architecture domains (business, data, application and technology) then translate strategy into design. In TOGAF terms, motivation sits in the Strategy and Motivation area at the start of the ADM cycle (see *Use of TOGAF and ArchiMate in SAF*). It holds the content that TOGAF produces before any architecture domain is designed, and that content keeps steering the architecture in all later phases.

ADM phase	Role of motivation in this phase	SAF sections
Preliminary	Establish the architecture capability: who the architecture serves and which rules guide decisions	Stakeholders, Architecture Principles
A. Architecture Vision	Explain why change is needed, what must be achieved and what value it brings	Drivers, Goals, Outcomes, Meaning, Value
Requirements Management	Turn goals into requirements and constraints, and keep them traceable through every phase	Requirements, Constraints
B, C, D. Core architectures	Justify design choices in business, application and technology architecture	Traceability from each design decision back to a goal and a driver
E to H. Implementation & Migration	Prioritise solutions and work packages; reassess when drivers change	Drivers and goals as criteria for prioritisation and change requests

Three consequences follow from this position.

- **Motivation comes first.** In TOGAF, no architecture domain is designed until stakeholders, drivers, goals and principles are agreed. The SAF follows the same order. Every later element must be traceable to the motivation.
- **Motivation stays active throughout the cycle.** In Requirements Management, at the centre of the ADM, goals are refined into requirements, and those requirements are checked in every phase. Motivation is the reference against which each architecture decision is tested, not a one-off introduction.
- **Motivation closes the loop.** In Phase H (Architecture Change Management), a new or changed driver, such as a new regulation or a new data source, triggers a change request. The cycle then returns to the Preliminary Phase or Phase A, and the motivation elements become the starting point of the next iteration.

The Strategy layer shares the Strategy and Motivation area with the Motivation layer. Motivation explains *why*. Strategy describes *what* the NSI must be able to do (capabilities) and *how* it creates value (value streams). Strategy is the bridge to Phase B, Business Architecture.

How NSIs use this chapter

The motivation elements in this chapter are a reference set. They describe internal and external forces and intentions that most NSIs share. Modelling the motivation layer is not mandatory, but the SAF strongly recommends it. Most NSIs have a motivation, but it is implicit: it lives in strategy documents, board decisions and the experience of staff, and is rarely connected to architecture decisions. Making it explicit, in SAF terms, helps an NSI in four ways.

- **It justifies investment.** If every programme can be traced to a driver and a goal, an investment proposal rests on shared reasoning, not on individual preference. Boards and ministries can see what a proposal responds to, and what is at stake if it is not funded. The same links set priorities: work that addresses several drivers, or contributes to a key outcome, comes before work that addresses none.
- **It makes decisions consistent.** Principles turn intentions into decision rules, so similar questions get similar answers, whichever team or architect is involved. When circumstances change, for example through new legislation, the documented links show which goals, principles and capabilities are affected, and the NSI can review those decisions instead of starting over.
- **It preserves knowledge.** Staff move on and projects end. A documented motivation keeps the reasoning behind past decisions available, so new staff can see why the architecture looks the way it does.

- **It supports collaboration between NSIs.** With the same concepts and identifiers, NSIs quickly see where their interests overlap. That makes it easier to start joint projects, share components and compare approaches. A documented motivation is also useful evidence in external assessments, such as peer reviews.

An NSI adopts, adapts or extends each SAF motivation element (see *Positioning the SAF in your own architecture* in the Reading guide). The Toolkit instrument named in the table below supports each step.

Toolkit (planned): Modernisation maturity scan (see *Planned instruments* in the Toolkit)

Motivation concepts in the SAF

SAF section	ArchiMate element	What it is	SAF content	Toolkit
Drivers	Driver	An external or internal condition that motivates an organisation to set goals and to change. A force, not a problem and not a solution.	Reference: SAF-D1 to SAF-D5	SAF-TK-M02
Goals	Goal	A statement of intent: the direction or end state an organisation wants to achieve. It says *what*, not *how* or *how much*.	Reference: SAF-G1 to SAF-G5	SAF-TK-M04
Meaning	Meaning	The knowledge or interpretation a concept has in a given context: *what does this signify?*	Reference: SAF-M1 to SAF-M4	SAF-TK-M09
Outcomes	Outcome	An end result that has been, or is intended to be, achieved: what will be different once a goal has been followed.	Reference: SAF-O1 to SAF-O6	SAF-TK-M05
Architecture Principles	Principle	A durable, general rule that guides how an organisation designs, builds and changes its information provision, for all decisions within its scope.	Reference: SAF-P01 to SAF-P11	SAF-TK-M06

SAF section	ArchiMate element	What it is	SAF content	Toolkit
Value	Value	The worth, usefulness or importance of something to a stakeholder: *what is it worth, and to whom?*	Reference: SAF-V1 to SAF-V6	SAF-TK-M09
Assessment	Assessment	The result of analysing a driver for one organisation: what the driver means for it today.	Defined by each NSI	SAF-TK-M03
Requirements	Requirement	A statement of need that a specific solution, process or platform must meet to contribute to a goal.	Defined by each NSI	SAF-TK-M07
Constraints	Constraint	A factor that limits how goals can be realised: the boundaries within which a solution must be built.	Defined by each NSI	SAF-TK-M08
Stakeholders	Stakeholder	An individual, group or organisation with an interest in the effects of the architecture **[R8]**, with concerns such as cost, risk, quality or compliance.	Reference: SAF-ST01 to SAF-ST07	SAF-TK-M01

The concepts form a chain. Stakeholders have concerns about drivers. An assessment records what a driver means for one NSI. Drivers motivate goals. Outcomes make goals measurable. Principles guide how goals are pursued, and requirements and constraints make them concrete for a solution. Meaning and value express why the result matters to stakeholders.

Relationships between the motivation elements

Element	Relationship	Example in the SAF
Stakeholders → Drivers	Stakeholders have concerns about drivers and often raise them	CIOs and CTOs (SAF-ST01) raise resource constraints (SAF-D5)
Assessment → Drivers	An assessment analyses a driver for one NSI	"Two thirds of our processing runs on platforms out of vendor support" assesses SAF-D5
Drivers → Goals	Drivers motivate goals	Resource constraints (SAF-D5) motivate cost efficiency (SAF-G1)

Element	Relationship	Example in the SAF
Drivers → Principles	Drivers explain why a principle exists	Trust and data protection (SAF-D3) explain privacy by design (SAF-P11)
Drivers → Constraints	Some drivers are the origin of constraints	SAF-D3 is the origin of data protection constraints
Assessment → Goals	Assessments influence which goals an NSI sets and how it prioritises them	A weakness on SAF-D5 makes modernisation (SAF-G3) a priority
Outcomes → Goals	Outcomes make goals measurable	Increased efficiency (SAF-O5) shows progress on cost efficiency (SAF-G1); improved interoperability (SAF-O3) on the harmonised reference framework (SAF-G2)
Assessment → Outcomes	A new assessment shows whether outcomes have been reached	The share of unsupported platforms falls after a modernisation programme (SAF-O1)
Principles → Goals	Principles guide how goals are pursued	Open-source participation (SAF-P01) guides cost efficiency (SAF-G1); loose coupling (SAF-P07) guides modernisation (SAF-G3)
Stakeholders → Goals	Stakeholders own goals	CIOs and CTOs (SAF-ST01) and NSIs (SAF-ST05) own modernisation (SAF-G3)
Requirements → Goals, Outcomes	Requirements make goals concrete for a solution and make outcomes measurable in it	SAF-G4 leads to a requirement for automated testing in the production pipeline; lineage recording contributes to SAF-O2
Requirements → Principles	Requirements apply principles to a specific solution	SAF-P09 becomes the requirement that no dataset is handed over without its minimum metadata **[R5]**
Constraints → Goals, Requirements, Principles	Constraints limit how goals can be realised, bound the solutions that can meet a requirement, and set the limits within which principles apply	A ban on public cloud for statistical microdata limits SAF-G3; a scalability requirement must be met on premises; SAF-P01's reuse–open source–buy–build order applies within public procurement rules
Stakeholders → Constraints	Stakeholders often own or impose constraints	A privacy officer applies data protection law; a ministry sets the budget
Outcomes → Value	Outcomes deliver value to stakeholders	Enhanced data quality (SAF-O2) delivers trust and legitimacy (SAF-V6); improved interoperability (SAF-O3) delivers cross-border comparability (SAF-V1)

Element	Relationship	Example in the SAF
Meaning → Value, Goals	Meaning explains how value is interpreted and frames how goals are understood	The SAF as common language (SAF-M4) explains why cross-border comparability (SAF-V1) is achievable; the SAF as bridge between strategy and technology (SAF-M1) underpins strategic alignment (SAF-V4); the SAF as foundation for transformation (SAF-M3) frames modernisation (SAF-G3)
Stakeholders → Meaning, Value	Meaning is shaped for and by stakeholders; stakeholders perceive value	IT and data professionals (SAF-ST04) read the SAF through its notation (SAF-M4); policy makers (SAF-ST03) value trust and legitimacy (SAF-V6); researchers (SAF-ST06) value innovation enablement (SAF-V2)
Stakeholders → Viewpoints	Stakeholder concerns determine which views are produced	Privacy officers need data-flow and protection views
Stakeholders → Business actors and roles	A stakeholder may also act in the architecture	Data providers are stakeholders and actors in VS-01
Meaning → Glossary	The meaning of individual terms is recorded in the glossary	The definition of "steady state"
Capabilities (Strategy)	Capabilities are developed in response to drivers, through goals, and realise goals and outcomes; a capability maturity assessment is a structured form of assessment	AI and technological acceleration (SAF-D2) drives investment in methodology (2.2.3), which supports data innovation (SAF-G4); metadata management (2.1.2) realises enhanced data quality (SAF-O2); a low score on 2.1.2
Value streams (Strategy)	Value streams show where a driver or goal affects the production chain, produce outcomes and create value stage by stage; each applies a subset of the principles	SAF-D1 affects data acquisition (VS-01); sustainable compliance (SAF-G5) is put into practice in VS-07; the steady-state handovers in VS-01 to VS-06 contribute to SAF-O3; VS-06 delivers released statistics to users; VS-07 enforces SAF-P09, P10 and P11 across the chain
Steady states (Steady-State Architecture)	Principles shape the design rules; quality gates make outcomes observable; gate criteria and minimum metadata work as reusable requirements	SAF-P07, P08, P09 and P11 underpin the design rules; lineage and metadata per state provide evidence for SAF-O2; the criteria for registering data in State #1

Elements defined by each NSI

The SAF does not define assessments, requirements or constraints, because they depend on the situation, the solution and the maturity of each NSI. It describes how they are used, and the Toolkit provides templates and illustrative examples.

Assessment. An assessment records what a driver means for one NSI: a strength, a weakness, an opportunity or a threat. A driver describes a force; an assessment describes how that force affects this NSI, today. Assessments are specific to one organisation ("two thirds of our processing runs on platforms out of vendor support"), based on evidence such as an application portfolio scan, an audit finding, a maturity

assessment or a staff survey (an opinion without evidence is not an assessment), and they change as the NSI acts on them, so they are reviewed regularly, for example once a year. A generic assessment would contradict the concept itself. Assessments are the link between the generic SAF drivers and the goals of an NSI. Without them, an NSI jumps from "the world is changing" straight to "we must modernise", and nobody can see why this particular goal has priority. An assessment also gives a baseline: if the next assessment of the same driver is better, the NSI can show progress. Assessments are produced in the *Assess* step of applying drivers, often as a SWOT analysis, and feed the *Respond* step, in which the NSI derives or confirms its goals. In TOGAF terms, they are part of Phase A and of the baseline description of each architecture domain.

Requirements. A goal says what the NSI wants to achieve; a requirement says what a specific solution, process or platform must do or provide to contribute to it. Requirements apply to a specific solution ("the new processing platform records the lineage of every registered dataset"), are testable in an architecture review or an acceptance test, and are derived, not invented: a requirement that cannot be traced to a goal, an outcome or a principle needs a justification. Principles and goals are deliberately general and only change a design when they are made concrete. Requirements connect the motivation layer to the architecture domains, and let an architecture board test whether a solution really contributes to the goals it claims to serve. In TOGAF, they are managed in Requirements Management and checked against each architecture decision. An NSI derives them from goals and outcomes, from principles, and from SAF reference content. The SAF already contains content that works as requirements: the quality gate and minimum metadata set of each steady state, which an NSI that adopts the steady-state model can use directly for the processes and platforms that produce and store each state, and the principle tables of each value stream.

Constraints. A requirement describes what a solution must do; a constraint describes the boundaries within which it must be done. Constraints are usually not chosen by the architect. They limit the solution space ("personal data may not be processed outside the EU" excludes certain cloud services, whatever their other merits), they come from outside the design (legislation, government policy, budget, staffing, existing contracts, technical dependencies), and they differ in how fixed they are: a legal constraint cannot be negotiated, a budget or policy constraint may change, so recording the source shows which ones can be challenged. Constraints that stay implicit cause late surprises in a procurement, a privacy assessment or a security review. Recording them early, and linking them to the goals they affect, keeps the architecture realistic and makes trade-offs visible: when a constraint blocks a goal, the board can accept that or challenge the constraint. In TOGAF, constraints are identified in Phase A and managed with requirements in Requirements Management. These are constraints on an NSI's own architecture, different from the *Constraints & Assumptions* of the SAF itself (SAF-C1, SAF-C2). Some apply to almost every NSI, such as data protection law and the statistical confidentiality required by national statistical law. *Definition of Scope* leaves the technology stack and organisational design to the NSI and its constraints, and SAF-P10 and SAF-P11 point to the legal and security frameworks that usually impose constraints.

Drivers

Drivers make the SAF a response to real pressures instead of an abstract ideal. With them, an NSI can explain to its board, its ministry or a peer review why change is needed, and what happens without it. They also help to set priorities: an investment that responds to several drivers deserves more attention than one that responds to none. "Rapid technological change" is a driver. "Our methods cannot use administrative data" is an assessment of that driver for one NSI. "Integrate new data sources into production" is a goal.

Drivers exist independently of the NSI: an NSI can respond to a driver, but cannot decide whether it exists. External drivers come from technology, legislation, society, politics or the economy; internal drivers from legacy systems, skills or organisational structure. Drivers are relatively stable and usually stay relevant for several years; short-lived events, such as an incident or a budget round, are recorded as assessments. The SAF drivers are common to the UNECE statistical community.

SAF drivers

ID	Driver	Type	Description	Architectural response in the SAF
SAF-D1	Evolving data ecosystems and partnerships	External	Official statistics move from isolated data collection to a wider data ecosystem. Surveys now sit next to administrative, commercial, platform and community data. This opens up new forms of collaboration, but makes NSIs dependent on actors, technologies and infrastructures they do not control. Open source is part of this ecosystem: a shared base for innovation, transparency and lower costs, developed with peers, academia and private partners.	Architectures for secure data sharing, standard interfaces and transparent governance, so that NSIs can take part in national data platforms and European data spaces, where production may span several environments and jurisdictions. Openness is combined with strong governance and clear accountability, so that collaboration does not compromise the integrity of official statistics.

ID	Driver	Type	Description	Architectural response in the SAF
SAF-D2	Artificial intelligence and technological acceleration	External	AI, automation and rapid technological progress change how data is created, processed and used. New tools can automate routine tasks, improve analysis and strengthen quality assurance, but raise the question of what relevant and trusted statistics mean when users rely on intelligent systems. NSIs must both use AI and make their outputs usable by AI: internally through automation, machine learning and advanced analytics within secure and transparent limits; externally through statistics that are machine-readable, machine-understandable and interoperable with the platforms of policy makers, researchers and citizens.	A risk-based approach: processes that demand methodological rigour and control stay tightly governed, less sensitive areas can be automated further. Modular, standards-based design lets an NSI adopt new technology at its own pace, while it stays fully accountable for quality, data protection and independence.

ID	Driver	Type	Description	Architectural response in the SAF
SAF-D3	Trust, scientific rigour and data protection	External	Trust is the basis of official statistics. Misinformation, privacy concerns and unregulated data providers put it under pressure. Compliance alone does not maintain public confidence; scientific rigour, transparency and clear accountability for the handling of data are also needed.	Trust and protection built into the architecture: governance and technical controls that protect confidentiality, keep methods sound and make processes traceable. Oversight is strict where sensitivity and quality demands are high; where risks are lower, more automation and shared solutions are possible.
SAF-D4	Uncertain and polarised global environment	External	Global tensions, political polarisation and shifting power relations affect how official statistics are demanded and perceived. Cross-border data flows require statistical systems to stay open and interoperable, without giving up national and institutional integrity.	A stable base for cooperation: interoperable architectures based on shared standards let NSIs exchange data and methods securely. Design principles that protect independence and transparency help NSIs stay credible and comparable, even when facts themselves are contested.

ID	Driver	Type	Description	Architectural response in the SAF
SAF-D5	Resource constraints and organisational agility	Internal and external	Many NSIs must do more with limited budgets, ageing systems and a shortage of specialist staff, while demand for timely, detailed and cross-domain output keeps growing. Meeting that demand requires efficiency and the ability to adapt quickly.	Flexibility and reuse as core principles: modular design, shared services and scalable solutions that reduce duplication. Clear governance of priorities lets NSIs shift capacity, modernise step by step and keep delivering quality output under financial and staffing pressure, without large reinvestment.

Toolkit: SAF-TK-M02 Driver recognition

Goals

Goals define what success looks like, for the SAF and for the NSIs that apply it. They give direction to the Strategy layer: capabilities and value streams are developed because they contribute to a goal. Without explicit goals, modernisation can turn into a series of technical upgrades that nobody can explain.

Goals are qualitative and directional: "modernise statistical systems" sets a direction, while a measurable result such as "retire all platforms out of vendor support by 2029" is an outcome. They are long-term, usually spanning several planning cycles and outliving individual programmes, and they only have meaning if a stakeholder recognises them as their intention. The SAF goals are shared by the UNECE statistical community and cover operational and strategic needs. Some focus on efficiency, modernisation and compliance, which keep an NSI sustainable. Others support innovation and collaboration, such as the use of new data sources or advanced analytics. Together they make sure that modernisation does not weaken the core values of official statistics: neutrality, accuracy and reliability.

SAF goals

ID	Goal	Description	Architectural direction in the SAF
SAF-G1	Cost efficiency and resource sharing	Budgets are limited and demand keeps growing, so NSIs must work as efficiently as possible. By sharing services, reusing solutions, improving interoperability and pooling resources, NSIs gain capabilities they could not build alone and avoid duplicate work. Efficiency then supports quality and long-term sustainability instead of competing with them.	Shared services, reusable components, and open-source components and community platforms where they improve transparency, collaboration and cost. Modular architectures and clear governance let NSIs adapt to new needs, shift capacity and modernise step by step, without large reinvestments.
SAF-G2	Harmonised reference framework	Without a shared architectural reference, NSIs build isolated solutions that reduce comparability and raise long-term costs. The SAF offers a common base of standards, patterns and practices that NSIs adapt to their own context while staying interoperable across borders and domains.	Common reference models and consistent design principles. Collaboration increasingly takes place on shared infrastructures, such as national data platforms and international data spaces, and harmonisation is a condition for taking part. Consistent principles help NSIs protect the integrity of official statistics and reuse shared and open solutions safely.
SAF-G3	Modernisation of statistical systems	Many NSIs still depend on fragmented legacy IT that slows down innovation and response. Moving to modular, service-based architectures that are secure, scalable and easier to maintain reduces technical debt, improves reliability and speeds up the delivery of new and better statistics.	Service-based design. Modernisation more often happens with several parties, from national infrastructures to cross-border data spaces, so the SAF sets principles for access control, data protection and accountability in those settings. A risk-based approach to automation: critical processes stay tightly governed to protect methodological rigour; less sensitive steps can be automated or shared.

ID	Goal	Description	Architectural direction in the SAF
SAF-G4	Support for data innovation	New data and methods are central to the future of official statistics, yet many NSIs struggle to introduce a new source or product: their production systems are built for predictability and stability, qualities that are necessary but slow down change. New sources, methods and products must enter production without loss of quality or consistency.	Architectures that combine robustness with flexibility. Versioning, validation and automated testing keep production reliable as it changes. Shared platforms and processes help prototypes move into operational use. Open source lets NSIs build on each other's work, supported by patterns for the safe reuse of open components.
SAF-G5	Sustainable compliance	Compliance with laws, standards and ethical principles underpins the trust in and legitimacy of official statistics. Statistical production is now part of a wider data and technology ecosystem, so compliance must extend beyond statistical standards, inside and outside the statistical domain.	GSBPM, GSIM and SDMX remain the basis for consistent processes, metadata management and data exchange, combined with relevant standards for data management, AI and cloud, so that NSIs stay interoperable with modern platforms. With consistent governance and open, verifiable technology, compliance is designed in from the start and becomes an enabler: innovation, data sharing and collaboration take place in controlled and auditable environments, with transparency, scientific rigour and privacy by design in every process.

Toolkit: SAF-TK-M04 Goal alignment

Meaning

Meaning answers the question *what does this signify?* Value answers a different question: *what is it worth, and to whom?* The SAF uses meaning at the level of the framework itself: the meaning elements describe how the statistical community should understand the SAF. Architecture frameworks are easily misread. IT staff may see a technical blueprint; managers may see administrative overhead. Explicit meaning elements set the intended interpretation, so that all stakeholders read the framework in the same way.

Two interpretations matter most. First, the SAF connects strategy and technology: every technology decision, from a new platform to a redesigned process, can be traced to a strategic objective. Next, the SAF

is a common language for the UNECE statistical community: shared models, practices and principles reduce duplicate work and make collaboration easier. This makes the SAF relevant beyond IT, to statisticians, policy makers and the citizens who depend on reliable statistics and information.

SAF meaning elements

ID	Meaning	Description
SAF-M1	Bridge between strategy and technology	The SAF links strategic objectives, such as relevance and efficiency, to technology choices. IT investments are justified by what they contribute to the mission.
SAF-M2	Catalyst for collaboration	The SAF is a common framework for the statistical community. It makes joint platforms, interoperable systems and harmonised standards possible, and strengthens what UNECE members can do together.
SAF-M3	Foundation for digital transformation	The SAF is more than a set of technical models. It is the basis for digital transformation in official statistics, without loss of methodological rigour or institutional trust.
SAF-M4	Common language and notation	Through TOGAF and ArchiMate, stakeholders share one language and notation. This reduces misunderstanding and raises architectural maturity across the community.

Toolkit: SAF-TK-M09 Value and meaning brief

Outcomes

Outcomes show stakeholders what adopting the framework returns. They connect long-term goals to concrete changes in capabilities, processes and systems, and make progress measurable, for one NSI and across the community. Outcomes are results, not activities: "introduce a metadata catalogue" is an activity, "every registered dataset has complete metadata and lineage" is an outcome. They are measurable, or at least observable, because an outcome without an indicator cannot be used to steer. Capabilities, value streams and solutions exist to produce them.

SAF outcomes are technical as well as societal. On the technical side, NSIs gain simpler processes, better metadata management and faster system development. On the societal side, the SAF supports trust, transparency and comparability, which informed policy and public accountability depend on. Outcomes also change over time: as an NSI matures, it sets new targets and adjusts its course. The indicative measures below are examples and the starting point of SAF-TK-M05.

SAF outcomes

ID	Outcome	Description	Indicative measures (illustrative)
SAF-O1	Accelerated modernisation	NSIs modernise faster and with less rework, because they start from shared reference models, patterns and building blocks instead of designing from scratch.	Lead time from decision to production for a new statistical product; share of solution designs that reuse SAF building blocks
SAF-O2	Enhanced data quality	Governance, metadata and lineage are part of the architecture. Statistical output therefore meets high standards of quality and reliability. Quality is designed in, not checked afterwards.	Share of datasets with complete metadata and lineage; revision rate of published figures
SAF-O3	Improved interoperability	NSIs reach higher interoperability across systems, domains and borders. This makes international collaboration easier, improves comparability and simplifies the use of external data sources.	Share of data exchanges using standard interfaces (e.g. SDMX); number of components shared between NSIs
SAF-O4	Improved trust and transparency	Openness, security and transparency in the architecture help to build and keep public trust. This strengthens the reputation of NSIs and the legitimacy of their output.	Results of public trust surveys; number of confidentiality or security incidents
SAF-O5	Increased efficiency	Reuse, harmonisation and standardisation reduce duplicate work. The result is lower IT costs, shorter development cycles and more efficient processing.	Number of redundant systems retired; processing time per production cycle; IT cost per statistical product
SAF-O6	Shared knowledge base	A community of practice across UNECE members exchanges lessons learned, reusable components and knowledge. This leads to continuous improvement at lower cost.	Number of reusable components and patterns published; number of NSIs contributing

Toolkit: SAF-TK-M05 Outcome and indicator sheet

Architecture Principles

Principles turn goals into decision rules. When an NSI chooses between technologies, solutions or designs, principles make the choice consistent and explainable, and they give an architecture board a shared basis for reviewing projects. They matter in official statistics because production runs in a closely connected environment, where a choice in one place affects many others. They prevent short-term convenience from undermining long-term sustainability, a real risk for NSIs with legacy systems and tight budgets. And they create accountability: when stakeholders know the principles behind a decision, investments and trade-offs are easier to explain. Principles are durable and rarely change.

The SAF follows the TOGAF template, which describes each principle by its name, its statement (the rule itself), its rationale (why the rule exists and what it brings) and its implications (what following it requires in cost, resources and activities). A principle is only useful if it is applied; a principle that never changes a decision is a slogan. Principles take effect only when they are part of governance: architecture reviews test compliance, and deviations follow a documented dispensation process. The Statistical Data Value Chain shows how each value stream applies the principles.

SAF architecture principles

Table 1: SAF architecture principles

ID	Principle	Statement	Rationale	Implications
SAF-P01	Actively participate in open-source communities to maintain collective control over statistical processes	Statistical organisations build statistical capabilities on shared open-source software, contribute to its communities, and release software they develop for statistical production as open source by default. Non-statistical processes use standard applications.	Critical statistical methods stay under the collective control of the official-statistics community rather than vendors, and open methods are transparent and reproducible. Sharing costs and contributing, not only consuming, keeps common components alive.	Contribution is planned and funded, improvements go back to the community project, and choices follow the order reuse, open source, buy, build, with building justified. Own components are released under an open licence unless security or confidentiality prevents it, and purchased software must support open standards, APIs and data export.
SAF-P02	Active support	Hardware and software have an active user community or vendor support. As a rule of thumb, the latest or second-to-last major version is used.	Business continuity requires fast and competent support when something breaks. Security requires that known vulnerabilities are fixed in time.	Support status is a selection criterion for every component. Components that approach end of support are flagged in the life cycle plan.
SAF-P03	Active life cycle management	Systems and components are kept up to date according to an agreed life cycle management (LCM) policy.	Without LCM, systems become outdated and vulnerable, and technical debt builds up. Paying off that debt later takes disproportionate effort and creates continuity risks.	Every component has an owner and a life cycle plan. LCM has a structural budget, not only when problems arise.

ID	Principle	Statement	Rationale	Implications
SAF-P04	Maximise the benefits for the organisation	IT decisions maximise the benefit for the organisation as a whole.	Decisions taken from an organisation-wide view have more long-term value than decisions taken from a departmental view. The principle stands for "service above self".	Local optimisations need justification against organisation-wide priorities. Shared solutions are preferred over departmental ones, even when they fit one department less well.
SAF-P05	IT and technology is everyone's business	All parts of the organisation take part in the IT decisions needed to reach organisational goals.	The users of IT are its main stakeholders. IT only fits the business when business experts and technical staff set goals and objectives together.	Business ownership of each information system is assigned explicitly. Architecture and change decisions are taken in joint business and IT forums.
SAF-P06	Business continuity	Work continues despite system or staff outages.	The more an NSI depends on its systems, the more their reliability matters. Hardware failures, disasters and data corruption must not stop statistical production or release.	Continuity requirements are part of every design. Critical processes have fallback arrangements, and recovery is tested regularly.
SAF-P07	Implement loosely coupled processes and systems	Processes and systems are built as independently as possible. They share functionality and logic through standard interfaces.	Loose coupling makes it easier to change or replace a single process or system. It also improves scalability and robustness, and it makes different security and privacy requirements easier to enforce.	Domains exchange data through defined interfaces, such as the steady-state handovers. Direct dependencies on another domain's internal data or logic are avoided.
SAF-P08	Data is shared property	Statistical data that adds value for internal or external users is shared.	Sharing enables reuse. In data collection, reuse reduces response burden. In processing, it avoids duplicate work on cleaning and analysing data.	Data is registered and findable in a catalogue. Before new data is collected, reuse of existing data is checked. Access is governed, not withheld by default.
SAF-P09	No data without metadata and classification	All shared data has metadata that describes its content and meaning as well as possible, and an AIC classification (availability, integrity, confidentiality).	Metadata makes data FAIR. An AIC classification sets the technical and organisational measures that laws, regulations and security standards require.	Metadata and classification are recorded when data is registered, not afterwards. Data without the required metadata is not handed over.

ID	Principle	Statement	Rationale	Implications
SAF-P10	Security by design	Security is an integral part of every solution. Security mechanisms are consistent across all layers of the architecture and scale from small to large objects.	Security added afterwards costs more, works less well and leaves gaps between layers.	Security requirements are set in the design phase of every solution and tested before production.
SAF-P11	Privacy by design	Privacy is an integral part of system architecture and business processes.	Privacy added afterwards conflicts with functionality and leaves risks. Built in from the start, it becomes part of the core functionality without limiting it.	Privacy impact is assessed in the design phase. Measures such as pseudonymisation, data minimisation and disclosure control are placed at defined points in the value chain.

Toolkit: SAF-TK-M06 Principle set and dispensation record

Value

Value makes the case for architecture in terms that stakeholders recognise. Architecture competes with other priorities for money and attention, and it only gets support when its long-term worth is clear. Value is always value *for someone*: the same outcome can be worth a lot to one stakeholder and little to another. An outcome is a result; value is what that result is worth to a stakeholder. A goal is an intention of the organisation; value is the reason a stakeholder cares whether the goal is reached. Meaning is how something is interpreted; value is how much it is worth.

The SAF creates value at several levels. For NSIs, it means lower risk, more agility and better alignment between IT and business. For international organisations, it means harmonisation and comparability. For citizens and policy makers, it means statistics they can rely on. Value also has a time dimension. Without a framework, modernisation tends to be fragmented, costly and short-lived. The SAF gives a stable base that reduces technical debt and makes investments last.

SAF value elements

ID	Value	Description
SAF-V1	Cross-border comparability	Statistics are worth more when they can be compared across borders. Common standards, models and processes make that comparison meaningful.
SAF-V2	Innovation enablement	The SAF lowers the barriers for AI, machine learning, real-time analytics and big data. NSIs can experiment safely and keep their methods sound.

ID	Value	Description
SAF-V3	Risk reduction	Compliance, security and governance are built into the architecture. This reduces the risk of system failures, data breaches and regulatory non-compliance, and protects the trust of citizens and stakeholders.
SAF-V4	Strategic alignment	IT initiatives contribute directly to the mission of producing official statistics. This improves governance, prioritisation and the impact of IT investments.
SAF-V5	Sustainability	Efficient, modern architectures with less technical debt give NSIs a clear path forward, without repeated and costly overhauls.
SAF-V6	Trust and legitimacy	Transparency, ethical practice and robust security strengthen the position of NSIs as authoritative providers of public data.

Toolkit: SAF-TK-M09 Value and meaning brief

Stakeholders

Architecture is never developed in isolation. Knowing the stakeholders makes clear whose concerns the architecture must address, who should take part in governance and decisions, and which views each audience needs. Involvement also builds support: people who take part in architecture decisions are more likely to back them.

A stakeholder is not the same as a business actor or role. An actor or role performs work in the architecture, for example a data steward who registers datasets. A stakeholder has an interest in the architecture, whether or not it performs work in it. The same person or organisation can be both. SAF stakeholders range from CIOs and technical experts within NSIs to international organisations, policy makers, researchers and society as a whole. Their interests differ, and the framework has to balance them. IT professionals turn SAF principles into systems. Policy makers care about the results those systems make possible. UNECE keeps the SAF harmonised and relevant across countries.

SAF stakeholders

ID	Stakeholder	Category	Key concerns	What the SAF offers
SAF-ST01	CIOs and CTOs of NSIs	Within the NSI	Strategic IT investment, alignment of IT with business goals, compliance	A framework to compare options, align IT with business goals and show compliance with international standards

ID	Stakeholder	Category	Key concerns	What the SAF offers
SAF-ST02	Citizens and society	Society	Trustworthy, transparent and modern statistics	Statistics that support democratic accountability and informed public debate
SAF-ST03	Government policy makers	Society	Timely, accurate and comparable data in accessible formats	Better statistical production and dissemination (indirect benefit)
SAF-ST04	IT and data professionals	Within the NSI	Common patterns, tools and practices; collaboration across institutions	Shared patterns and practices that save effort and shorten the learning curve
SAF-ST05	National Statistical Institutes (NSIs)	Statistical community	Structured guidance for enterprise architecture; efficiency; quality	Guidance tailored to official statistics and aligned with international practice
SAF-ST06	Researchers and academia	Society	Access to high-quality data and metadata for research	Systems designed for openness, data sharing and rich metadata
SAF-ST07	UNECE and other international bodies	Statistical community	Relevance, coherence and adoption of the framework	A custodial role for UNECE; alignment with the modernisation work of Eurostat and the UN Statistics Division

Toolkit: SAF-TK-M01 Stakeholder map